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The method of the celestial unknown

↳ in East Asian mathematics in Mathematics in China

Karine Carole Chemla

Britannica Editors

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Table of Contents

Li Ye's book also contains a method, unknown to Qin Jiushao, that seems to have flourished in North China for some decades before Li completed "Sea Mirror of Circle Measurements." This method explains how to use polynomial arithmetic to find equations to solve a problem. Li's book is the oldest surviving work that explains this method, but it was probably not the first to deal with it. In this book polynomials are also arranged according to a positional notation. Thus, $x^2 - 3x + 5 + 7/x^2$ is represented as

7

0

5.

-3

1

A character is added next to the 5 (replaced by a dot on the image) to indicate that it is a constant term. The location of the coefficient indicates the power of the indeterminate with which it is associated. This indeterminate is called "the celestial unknown."

Research continued on these topics for several decades, as can be seen from the completion in 1299 of *Suanxue qimeng* ("Introduction to Mathematical Science") by Zhu Shijie, which devotes some problems to presenting the "procedure of the celestial unknown." Moreover, it is known that some mathematicians used this representation for polynomials in two or three unknowns; however, their writings are lost. In his second surviving book, *Siyuan yujian* ("Precious Mirror of the Four Elements"), Zhu made use of four unknowns. Starting from the centre of the counting board

Cite

the indeterminates or too many mixed terms occurred, however, he had to use tricks that were in conflict with the principles of the place-value notation. In problems where there was one unknown, he had to use a method of elimination of a common unknown between equations.

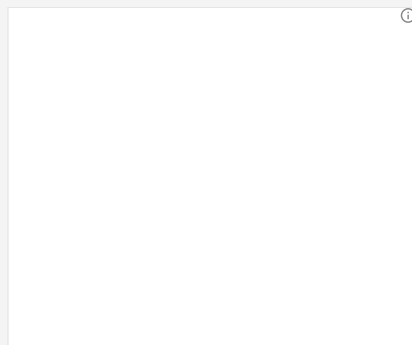


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Chinese remainder theorem

Qin Jiushao's book also contains algorithms for the general congruence problem, an example of which was given in Sunzi's 5th-century treatise, where its solution was too obscure to be understood. This problem amounts to determining a number, the remainders of which are known when it is divided by given numbers (called moduli). There is no extant work between Sunzi's treatise and Qin's book of 1247 that reveals how this algorithm was elaborated. Such problems seem to have been worked out because of calendrical computation. Qin introduced his discussion by saying that his goal was to clarify several procedures used by astronomers who were applying them without understanding them. His solution is known today as the [Chinese remainder theorem](#). He dealt with the case when moduli are relatively prime, and he then reduced the case when they are not by first eliminating common factors. The first case is easily solved when x can be found that satisfies the congruence $xa \equiv 1 \pmod{b}$, a and b being two given relatively prime numbers (suppose $a < b$). Qin gave an algorithm for this, using a sequence of quotients in searching for the greatest common divisor of a and b , which is also the sequence of convergents for the continued fraction for b/a . Having them, he was then able to compute x .



Fall into oblivion, 14th–16th centuries

Little is known about what happened to Chinese mathematics after Zhu Shijie, but surviving books from the following centuries attest to a progressive loss of the great achievements of the Song-Yuan period. In the 16th century a mathematician's comments on Li Ye's "Sea Mirror of Circle Measurements" show that the method of the celestial unknown was no longer understood. By the 17th century it seems to have been completely forgotten. Rods were then no longer used as a counting tool, so perhaps Chinese algebraic place-value notations, deprived of the instrument on which they were based, could not be understood.

On the other hand, there was a rapid diffusion of the [abacus](#), for which many books were written. One of them, the *Suanfa tongzong* ("Systematic Treatise on Mathematics") by Cheng Dawei (1592), had a special significance. In addition to its detailed treatment of arithmetic on the abacus, it provided a summa of mathematical knowledge assembled by the author after 20 years of bibliographic research. Re-edited several times through the 19th century, the "Systematic

Treatise” was the main source—and still is an important source—available to scholars in China, and more generally in East Asia, concerning mathematics as it developed in China.

When European missionaries arrived in China at the end of the 16th century, they found people interested in science (so that the missionaries were accepted in China because of their scientific knowledge) but unaware of their own past in mathematics. An era of translations of Western works then started, the first six books of [Euclid](#)’s *Elements* being translated by the Jesuit [Matteo Ricci](#) and Xu Guangqi in 1607. In parallel to this process of translation, Chinese scholars attempted to find ancient books, to understand them, and to synthesize the Chinese and Western traditions. In the 18th century, with the help of Western algebra, [Mei Juecheng](#) deciphered the ancient texts dealing with the method of the celestial unknown. This triggered a renewed search for ancient Chinese sources and attempts to revive mathematical research with traditional Chinese methods.



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Mathematics in Japan

The introduction of Chinese books

Very little is known about Japanese mathematics before the 17th century. Beginning in the 7th century, at first only indirectly by way of Korea, there was a flow of Chinese science to Japan. For example, the “[Ten Classics of Mathematics](#)” was introduced, along with counting rods, probably by the 8th century. Yet no Japanese book dealing with mathematics survives from before the end of the 16th century. At that time another phase of importation began: the [abacus](#) and Cheng Dawei’s “Systematic Treatise on Mathematics” became known in Japan, though they did not supplant the use of counting rods. Moreover, many books were brought from Korea, and perhaps in that way two Chinese books, *Yang Hui suanfa* (1275; “Yang Hui’s Methods of Mathematics”) and Zhu Shijie’s “Introduction to Mathematical Science,” arrived in Japan. In those books, Japanese scholars could find algorithms for solving systems of simultaneous linear equations and for searching for the root of an equation according to methods used in China in the 13th century; they could also find applications of the method of the celestial unknown (although these were not immediately understood). In addition, books on calendrical computations, which also contained mathematical knowledge, were imported. As a result of such infusions, Chinese mathematics greatly influenced the development of Japanese mathematics (for example, its algebraic orientation) and defined the context in which the Japanese tradition later opened to European mathematics.



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At the beginning of the Tokugawa period (1603–1867), contacts with foreigners were limited to trade with Chinese and Dutch ships through the port of Nagasaki. Some Chinese books, which by then may have contained Western knowledge, as well as Dutch books entered Japan secretly, but it is difficult to state how much, or what kind of, mathematical knowledge entered through that channel.

The elaboration of Chinese methods

Although not the first mathematical book written in Japan, *Jingoki* (“Inalterable Treatise”), published in 1627 by Yoshida Mitsuyoshi, seems to be the first book that played an important role in the emerging Japanese tradition. Inspired by the Chinese text “Systematic Treatise on Mathematics,” whose importance is stressed above, it described in Japanese the use of the *soroban*, an improvement of the Chinese abacus, and introduced some Chinese knowledge. Its many editions contributed to popularizing mathematics because most of the works on mathematics in Japan were written in Chinese and could not be widely read. In its enlarged edition of 1641, *Jingoki* introduced the method of performing computations with counting rods, which by then were no longer used in China. Moreover, inspired by his Chinese source, Yoshida added “difficult problems” that he left without solutions and recommended be posed to mathematicians. This initiated a tradition of challenges, reminiscent of those that took place in Europe during the Renaissance, that strongly stimulated the development of mathematics in Japan. In this context, mathematicians in the 1650s, relying on counting-rod computations and looking for new methods of solution, began to decipher the original methods of Chinese algebra—hinted at in the 1658 Japanese reprint of “Introduction to Mathematical Science”—which enabled them to advance beyond the classics. This contrasts with the situation in China, where the original methods could be understood only after the introduction of Western algebra.

Various Japanese authors disseminated traditional Chinese methods for the solution of problems.

Sawaguchi Kazuyuki's *Kokon sanpoki* (1671; "Ancient and Modern Mathematics") that "erroneous" problems could have more than one solution (in other words, equations have more than one root), but he left unanswered difficult problems involving simultaneous equations of the n th degree. Equations for their solution were published in 1674 by [Seki Takakazu](#), now considered to be the founder of the Japanese tradition of mathematics, or *wasan*. Seki founded what became the most important school of mathematics in Japan. (At this time, mathematics was widely practiced in Japan as a leisure activity.) As in other schools, disciples had to keep the school methods secret, and only the best among them knew most of these methods. Only slowly did they publish their secrets, which hindered the free circulation of ideas and which makes any attribution very difficult.

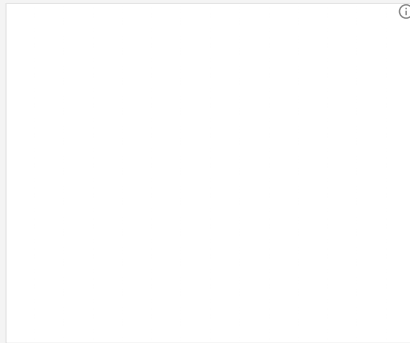
Explanations of how to use Seki's equations to derive Sawaguchi's problems were published in 1685 by one of Seki's disciples, [Takebe Katahiro](#). Seki had designed for this purpose a "literal" written algebra using [characters](#), thus liberating mathematicians from counting rods. He kept for equations the positional notation with respect to one unknown, the coefficients being expressed in terms of numbers, parameters, or other unknowns. In establishing equations among several unknowns for the solution of a problem, he had to introduce procedures equivalent to computations of determinants in order to eliminate unknowns between simultaneous equations. Further research elaborated these procedures.

Seki devised a classification of problems that amounted to a classification of equations, which took into consideration negative roots and multiple roots, the existence of which had been noticed by Sawaguchi; for this purpose he adapted the Chinese [algorithms](#) from the 13th century. Seki and his disciples thus improved upon Chinese methods in many ways, opening new directions for the development of mathematics in Japan—as, for example, in their work on [infinite series](#), the subject of research by contemporary European scientists as well.



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Yang Hui

Chinese mathematician

Also known as: *Yang Qianguang*

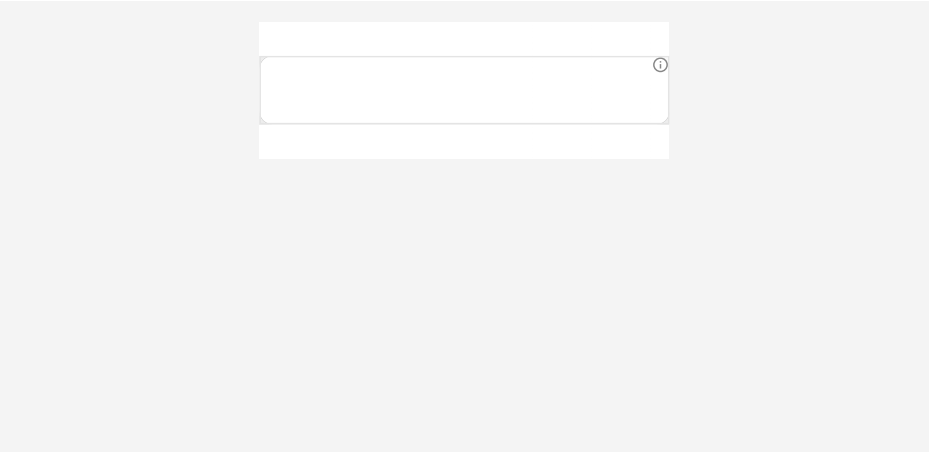
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Table of Contents

Yang Hui (flourished c. 1261–75, Qiantang, Zhejiang province, China) was a mathematician active in the great flowering of Chinese mathematics during the Southern Song dynasty.



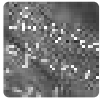
Although practically nothing is known about the life of Yang, his books are among the few contemporary Chinese mathematics works to survive. A remark in the preface to one of his treatises indicates that he was a mandarin (scholar-official).

Yang’s works are mentioned in the *Wenyan ge shumu* (1441; “Catalogue of the Books of the Ming Imperial Library”) but were long thought to be irreparably lost. Ruan Yuan, compiler of *Chou ren zhuan* (1799; “Biographies of Mathematicians and Astronomers”), first found fragments of Yang’s *Xiangjie jiuzhang suanfa* (1261; “A Detailed Analysis of the Nine Chapters on the Mathematical Procedures”) in a handwritten copy of an imperial Ming dynasty encyclopaedia, and he later discovered in Suzhou a Song dynasty edition of *Yang Hui suanfa* (1275; “Yang Hui’s Mathematical Methods”). The latter contains three treatises, *Chengchu tongbian benmo* (1274; “Fundament and Periphery for Continuity and Change in Multiplication and Division”), *Tianmu bilei chengchu jiefu* (1275; “Quick Methods for Multiplication and Division in Surveying and Analogous Categories”), and *Xu gu zhai qi suanfa* (1275; “Selection of Strange Mathematical Methods in Continuation of Antiquity”). A collected edition (1378) of these works was transmitted farther to the east, where it was particularly influential. In Korea it was reprinted during the reign of Sejong in 1433, and it was copied again by the Japanese mathematician Seki Takakazu (c. 1640–1708). Of another work, *Riyong suanfa* (1262; “Mathematical Methods for Daily Use”), only the preface and a few problems are known.

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Numbers and Mathematics



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Yang’s *Jiuzhang suan fa zuan lei* (c. 1275; “Reclassification of the Mathematical Procedures in the Nine Chapters”)—a compilation and reclassification, with further explanations, of the problems from the *Han dynasty* classic and its commentaries, *Jiuzhang suanshu* (c. 100 BC–AD 50; *Nine Chapters on the Mathematical Procedures*)—contains the oldest representation of what is known in the West as *Blaise Pascal’s* triangle (see the figure; see also *binomial theorem*). In the preface Yang asserts that he copied it from an older explication, *Huangdi jiuzhang suanfa* (“Yellow Emperor’s Nine Chapters on Mathematical Methods”) by *Jia Xian* (flourished c. 1050).



Blaise Pascal first described his triangle for generating the coefficients of a binomial expansion in 1665. The Ch...[\(more\)](#)



Yang’s “Mathematical Methods” was compiled with a pedagogic perspective. In the beginning of his book, he gives recommendations for the study of mathematics: start from the multiplication table, called “9 9 81” in the Chinese tradition, then study the positions for layout of numerals and the multiplication algorithms for higher numbers. In his collection he also describes in detail a geometric method for solving quadratic equations. A variety of *magic squares* can be found in “Strange Mathematical Methods,” including a 10-by-10 square such that each vertical and horizontal line of numbers adds to 505.

Quick Facts

- Literary name:** Qianguang
- Flourished:** c. 1261–75, Qiantang, Zhejiang province, China
- Flourished:** 1261 - 1275 • China
- Notable Works:** “Jiuzhang suan fa zuan lei” • “Xiangjie jiuzhang suanfa” • “Xu gu zhai qi suan fa” • “...[\(Show more\)](#)

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Takebe Katahiro

Japanese mathematician

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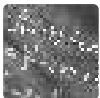
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☰ Table of Contents

Takebe Katahiro (born 1664, Edo [now Tokyo], Japan—died 1739, Edo) was a Japanese mathematician of the *wasan* (“Japanese calculation”) tradition (see [mathematics](#), [East Asian: Japan in the 17th century](#)) who extended and disseminated the mathematical research of his teacher [Seki Takakazu](#) (c. 1640–1708).

Takebe’s career was one of the most prestigious that a *wasan* mathematician ever experienced. He served successively two [shoguns](#), Tokugawa Ienobu (reigned 1709–12; see [Tokugawa period](#)), initially lord of [Kōfu](#), whom he escorted all along his rise to the supreme position, and [Tokugawa Yoshimune](#) (reigned 1716–45), an enlightened sovereign who gave a significant impulse to scientific research in Japan by encouraging scholars of various fields and by showing a personal interest in astronomy and calendar reform.

Takebe Katahiro became a pupil of Seki at the age of 13 and, together with his brother Kataaki, remained with him until his death in 1708. The brothers did their utmost to spread Seki’s work, to make it easier to understand, and to defend it against detractors. They were the main craftsmen of Seki’s project (launched 1683) to record mathematical knowledge in an encyclopaedia. The *Taisei sankei* (“Comprehensive Classic of Mathematics”), in 20 volumes, was finally completed by Takebe Kataaki in 1710. It gives a good picture of Seki’s skill at reformulating problems, as well as Takebe Katahiro’s ability to correct, perfect, and extend his master’s intuitions.



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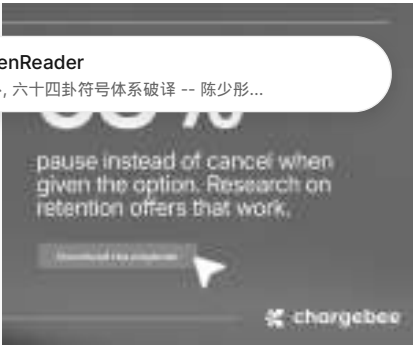
Numbers and Mathematics

The 1720s were Takebe’s most creative period. In his *Tetsujutsu sankei* (1722; “Art of Assembling”), a philosophical as well as a mathematical work, he explained what he regarded as the fundamental features of mathematical research. He distinguished two ways of solving a



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Died: 1739, Edo (aged 75)

Notable Works: “Tetsujutsu sankei”

Subjects Of Study: infinite series

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Annick Horiuchi

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Also known as: math

 Jeremy John Gray • All Britannica Editors June 6, 2026 • History



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Table of Contents

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mathematics, the science of structure, order, and relation that has evolved from elemental practices of counting, measuring, and describing the shapes of objects. It deals with reasoning and quantitative calculation, and its development has involved an increasing idealization and abstraction of its subject matter. Since the 17th century, mathematics has been an indispensable adjunct to the physical sciences and technology, and in more recent times it has assumed a similar role in the quantitative aspects of the life sciences.



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In many cultures—under the stimulus of the needs of practical pursuits, such as commerce and agriculture—mathematics has developed far beyond basic counting. This growth has been greatest in societies complex enough to sustain these activities and to provide leisure for contemplation and the opportunity to build on the achievements of earlier mathematicians.

All mathematical systems (for example, Euclidean geometry) are combinations of sets of axioms and of theorems that can be logically deduced from the axioms. Inquiries into the logical and philosophical basis of mathematics reduce to questions of whether the axioms of a given system ensure its completeness and its consistency. For full treatment of this aspect, see mathematics, foundations of.

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This article offers a history of mathematics from ancient times to the present. As a consequence of the exponential growth of science, most mathematics has developed since the 15th century CE, and it is a historical fact that, from the 15th century to the late 20th century, new developments in mathematics were largely concentrated in Europe and North America. For these reasons, the bulk of this article is devoted to European developments since 1500.



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This does not mean, however, that developments elsewhere have been unimportant. Indeed, to understand the history of mathematics in Europe, it is necessary to know its history at least in ancient Mesopotamia and Egypt, in ancient Greece, and in Islamic civilization from the 9th to the 15th century. The way in which these civilizations influenced one another and the important direct contributions Greece and Islam made to later developments are discussed in the first parts of this article.



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Fun Facts of Measurement & Math

India's contributions to the development of contemporary mathematics were made through the considerable influence of Indian achievements on Islamic mathematics during its formative years. A separate article, [South Asian mathematics](#), focuses on the early history of mathematics in the [Indian subcontinent](#) and the development there of the modern decimal place-value [numeral system](#). The article [East Asian mathematics](#) covers the mostly independent development of mathematics in China, Japan, Korea, and Vietnam.

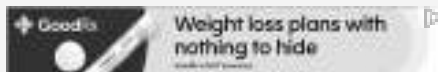
The substantive branches of mathematics are treated in several articles. See [algebra](#); [analysis](#); [arithmetic](#); [combinatorics](#); [game theory](#); [geometry](#); [number theory](#); [numerical analysis](#); [optimization](#); [probability theory](#); [set theory](#); [statistics](#); [trigonometry](#).

Key People: Luca Pacioli • Farkas Bolyai • Isaac Newton • Bertrand Russell • Alfred North Whitehead

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Ancient mathematical sources

It is important to be aware of the character of the sources for the study of the history of mathematics. The history of Mesopotamian and Egyptian mathematics is based on the extant original documents written by scribes. Although in the case of Egypt these documents are few, they are all of a type and leave little doubt that Egyptian mathematics was, on the whole, elementary and profoundly practical in its orientation. For Mesopotamian mathematics, on the other hand, there are a large number of clay tablets, which reveal mathematical achievements of a much higher order than those of the Egyptians. The tablets indicate that the Mesopotamians had a great deal of remarkable mathematical knowledge, although they offer no evidence that this knowledge was organized into a deductive system. Future research may reveal more about the early development of mathematics in Mesopotamia or about its influence on Greek mathematics, but it seems likely that this picture of Mesopotamian mathematics will stand.

From the period before Alexander the Great, no Greek mathematical documents have been preserved except for fragmentary paraphrases, and, even for the subsequent period, it is well to remember that the oldest copies of Euclid's *Elements* are in Byzantine manuscripts dating from the 10th century CE. This stands in complete contrast to the situation described above for Egyptian and Babylonian documents. Although, in general outline, the present account of Greek mathematics is secure, in such important matters as the origin of the axiomatic method, the pre-Euclidean theory of ratios, and the discovery of the conic sections, historians have given competing accounts based on fragmentary texts, quotations of early writings culled from nonmathematical sources, and a considerable amount of conjecture.

Many important treatises from the early period of Islamic mathematics have not survived or have survived only in Latin translations, so that there are still many unanswered questions about the relationship between early Islamic mathematics and the mathematics of Greece and India. In addition, the amount of surviving material from later centuries is so large in comparison with that which has been studied that it is not yet possible to offer any sure judgment of what later Islamic mathematics did not contain, and therefore it is not yet possible to evaluate with any assurance what was original in European mathematics from the 11th to the 15th century.

In modern times the invention of printing has largely solved the problem of obtaining secure texts and has allowed historians of mathematics to concentrate their editorial efforts on the correspondence or the unpublished works of mathematicians. However, the exponential growth of mathematics means that, for the period from the 19th century on, historians are able to treat only the major figures in any detail. In addition, there is, as the period gets nearer the present, the problem of perspective. Mathematics, like any other human activity, has its fashions, and the nearer one is to a given period, the more likely these fashions will look like the wave of the future. For this reason, the present article makes no attempt to assess the most recent developments in the subject.

John L. Berggren

Mathematics in ancient Mesopotamia

Until the 1920s it was commonly supposed that mathematics had its birth among the ancient Greeks. What was known of earlier traditions, such as the Egyptian as represented by the papyrus (edited for the first time only in 1877), offered at best a meagre precedent. This impression gave way to a very different view as historians succeeded in deciphering and interpreting the technical materials from ancient Mesopotamia.

Owing to the durability of the Mesopotamian scribes' clay tablets, the surviving evidence of this culture is substantial. Existing specimens of mathematics represent all the major eras—the Sumerian kingdoms of the 3rd millennium BCE, the Akkadian and Babylonian regimes (2nd millennium), and the empires of the Assyrians (early 1st millennium), Persians (6th through 4th century BCE), and Greeks (3rd century BCE to 1st century CE). The level of competence was already high as early as the Old Babylonian dynasty, the time of the lawgiver-king Hammurabi (c. 18th century BCE), but after that there were few notable advances. The application of mathematics to astronomy, however, flourished during the Persian and Seleucid (Greek) periods.

The numeral system and arithmetic operations

Unlike the Egyptians, the mathematicians of the Old Babylonian period went far beyond the immediate challenges of their official accounting duties. For example, they introduced a versatile numeral system, which, like the modern system, exploited the notion of place value, and they developed computational methods that took advantage of this means of expressing numbers; they solved linear and quadratic problems by methods much like those now used in algebra; their success with the study of what are now called Pythagorean number triples was a remarkable feat in number theory. The scribes who made such discoveries must have believed mathematics to be worthy of study in its own right, not just as a practical tool.

The older Sumerian system of numerals followed an additive decimal (base-10) principle similar to that of the Egyptians. But the Old Babylonian system converted this into a place-value system with the base of 60 (sexagesimal). The reasons for the choice of 60 are obscure, but one good mathematical reason might have been the existence of so many divisors (2, 3, 4, and 5, and some multiples) of the base, which would have greatly facilitated the operation of division. For numbers from 1 to 59, the symbols

𐎶
for 1 and
𐎵
for 10 were combined in the simple additive manner (e.g.,

𐎶𐎵𐎶𐎵𐎶𐎵
represented 32). But to express larger values, the Babylonians applied the concept of place value. For example, 60 was written as

𐎶
, 70 as

𐎶𐎵
, 80 as

𐎶𐎵𐎶
, and so on. In fact,

𐎶
could represent any power of 60. The context determined which power was intended. By the 3rd century BCE, the Babylonians appear to have developed a placeholder symbol that functioned as a zero, but its precise meaning and use is still uncertain. Furthermore, they had no mark to separate numbers into integral and fractional parts (as with the modern decimal point). Thus, the



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three-place numeral 3 7 30 could represent $3\frac{1}{8}$ (i.e., $3 + 7/60 + 30/60^2$), $187\frac{1}{2}$ (i.e., $3 \times 60 + 7 + 30/60$), 11,250 (i.e., $3 \times 60^2 + 7 \times 60 + 30$), or a multiple of these numbers by



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The four arithmetic operations were performed in the same way as in the modern decimal system, except that carrying occurred whenever a sum reached 60 rather than 10. Multiplication was facilitated by means of tables; one typical tablet lists the multiples of a number by 1, 2, 3, ..., 19, 20, 30, 40, and 50. To multiply two numbers several places long, the scribe first broke the problem down into several multiplications, each by a one-place number, and then looked up the value of each product in the appropriate tables. He found the answer to the problem by adding up these intermediate results. These tables also assisted in division, for the values that head them were all reciprocals of regular numbers.

Regular numbers are those whose prime factors divide the base; the reciprocals of such numbers thus have only a finite number of places (by contrast, the reciprocals of nonregular numbers produce an infinitely repeating numeral). In base 10, for example, only numbers with factors of 2 and 5 (e.g., 8 or 50) are regular, and the reciprocals ($1/8 = 0.125$, $1/50 = 0.02$) have finite expressions; but the reciprocals of other numbers (such as 3 and 7) repeat infinitely

$$0.\overline{3}$$

and

$$0.\overline{142857}$$

, respectively, where the bar indicates the digits that continually repeat). In base 60, only numbers with factors of 2, 3, and 5 are regular; for example, 6 and 54 are regular, so that their reciprocals (10 and 1 6 40) are finite. The entries in the multiplication table for 1 6 40 are thus simultaneously multiples of its reciprocal $1/54$. To divide a number by any regular number, then, one can consult the table of multiples for its reciprocal.

An interesting tablet in the collection of Yale University shows a square with its diagonals. On one side is written "30," under one diagonal "42 25 35," and right along the same diagonal "1 24 51 10" (i.e., $1 + 24/60 + 51/60^2 + 10/60^3$). This third number is the correct value of $\sqrt{2}$ to four sexagesimal places (equivalent in the decimal system to 1.414213..., which is too low by only 1 in the seventh place), while the second number is the product of the third number and the first and so gives the length of the diagonal when the side is 30. The scribe thus appears to have known an equivalent of the familiar long method of finding square roots. An additional element of sophistication is that by choosing 30 (that is, $1/2$) for the side, the scribe obtained as the diagonal the reciprocal of the value of $\sqrt{2}$ (since $\sqrt{2}/2 = 1/\sqrt{2}$), a result useful for purposes of division.



Babylonian mathematical tablet